NYC YELLOW TAXI TRIP DATA ANALYSIS

Final Capstone Project Report

Dataset: yellow_tripdata_2025-01.parquet
Period: January 2025
Tools: Python, NumPy, Pandas, Matplotlib, Seaborn, Jupyter Notebook

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1. Project Objective

The objective of this project is to analyze January 2025 NYC Yellow Taxi trip data and convert the raw trip records into clear and defensible business and operational insights.

The analysis focuses on:

- Data understanding
- Data quality assessment
- Data cleaning
- Feature engineering
- Statistical analysis
- Exploratory Data Analysis (EDA)
- Data visualization
- Business questions
- Key findings
- Business and operational insights
- Limitations
- Final conclusion

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2. Problem Statement

The NYC Yellow Taxi dataset contains information about taxi trips, including pickup and drop-off times, passenger count, trip distance, rate codes, pickup and drop-off locations, vendors, and payment types.

The purpose of this project is to analyze the dataset and identify meaningful patterns in taxi demand, trip characteristics, passenger behavior, locations, routes, payment types, vendors, and rate codes.

The project uses Python with NumPy, Pandas, Matplotlib, and Seaborn to perform data cleaning, transformation, statistical analysis, EDA, visualization, and interpretation.

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3. Dataset Description

Dataset File: yellow_tripdata_2025-01.parquet

Dataset Type: NYC Yellow Taxi trip records

Period: January 2025

Original Dataset Size: 3,475,226 rows and 20 columns

Important Columns

- VendorID
- tpep_pickup_datetime
- tpep_dropoff_datetime
- passenger_count
- trip_distance
- RatecodeID
- store_and_fwd_flag
- PULocationID
- DOLocationID
- payment_type
- fare_amount
- tip_amount
- total_amount
- congestion_surcharge
- Airport_fee
- cbd_congestion_fee

Important Dataset Limitation

This is a trip-level operational dataset. The analysis does not make unsupported claims about profit, driver earnings, customer spending, or customer-level behavior.

PULocationID and DOLocationID represent Taxi Zone IDs. They are not treated as neighborhood names because a separate Taxi Zone lookup table was not used.

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4. Tools and Technologies

Python

Used as the main programming language.

NumPy

Used for numerical and statistical calculations such as:

- Mean
- Median
- Minimum
- Maximum
- Variance
- Standard deviation
- Percentiles

Pandas

Used for:

- Loading data
- Data exploration
- Data cleaning
- Data transformation
- Feature engineering
- Grouping
- Aggregation

Matplotlib

Used for creating core visualizations.

Seaborn

Used for statistical and categorical visualizations.

Jupyter Notebook

Used to develop, execute, document, and present the analysis.

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5. Project Workflow

The project followed the workflow:

Raw Taxi Data
↓
Understand Dataset
↓
Data Quality Assessment
↓
Data Cleaning
↓
Feature Engineering
↓
Statistical Analysis
↓
Exploratory Data Analysis
↓
Visualization
↓
Business Questions
↓
Insights
↓
Final Conclusion

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6. Initial Data Exploration

The dataset was loaded into a Pandas DataFrame and initially explored to understand its structure.

The following were examined:

- First five records
- Last five records
- Number of rows
- Number of columns

- Column names
- Data types
- Unique values
- Missing values
- Statistical summary

The original dataset contained:

3,475,226 rows and 20 columns.

The pickup and drop-off datetime columns were already stored in an appropriate datetime format.

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7. Data Quality Assessment

The dataset was checked for:

- Missing values
- Duplicate records
- Invalid values
- Zero values
- Suspicious trip distances
- Suspicious passenger counts
- Invalid timestamps
- Zero-duration trips
- Extremely long trip durations
- Extremely large trip distances
- Unusual categorical values

Missing Values

There were **540,149 missing records** in several columns, including:

- passenger_count
- congestion_surcharge
- store_and_fwd_flag

- RatecodeID
- Airport_fee

These missing values occurred together.

The records were retained because the remaining information in those records was still useful for analysis.

Duplicate Records

No completely duplicated rows were found.

Duplicate records: 0

Trip Distance

Trip distance was checked for:

- Negative values
- Zero values
- Extremely large values
- Values above 100 miles
- Values above 1,000 miles

No negative trip distances were found.

Zero-distance trips were investigated but were not automatically removed.

Extreme distance values were also investigated and retained rather than automatically deleted.

Trip Duration

Trip duration was calculated using pickup and drop-off timestamps.

There were **124 records where the drop-off time occurred before the pickup time**.

These records were considered logically invalid.

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8. Data Cleaning

Trip duration was calculated using:

Drop-off Time − Pickup Time

Records with negative trip duration were removed because the timestamp sequence was logically impossible.

Cleaning Summary

Cleaning Metric	Value
Original Records	3,475,226
Records Removed	124
Final Cleaned Records	3,475,102
Duplicate Records	0

Cleaning Decision

The project did not automatically remove every unusual record.

The following were investigated and retained where there was not enough evidence to classify them as invalid:

- Zero-distance trips
- Zero-duration trips
- Extremely large distances
- Very long durations
- IQR outliers
- Extreme calculated speeds
- Missing-value records

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9. Feature Engineering

New columns were created to support the analysis.

Trip Duration

trip_duration_minutes

Calculated from pickup and drop-off timestamps.

Pickup Date

pickup_date

Extracted from the pickup datetime.

Pickup Hour

pickup_hour

Extracted from the pickup datetime to analyze demand by hour.

Day of Week

day_of_week

Extracted from the pickup datetime.

Weekend Indicator

is_weekend

Used to identify Saturday and Sunday trips.

Average Speed

average_speed_mph

Calculated using trip distance and trip duration for trips with positive duration.

Distance Category

Trips were categorized as:

- Short
- Medium
- Long

Duration Category

Trips were categorized as:

- Very Short
- Short
- Medium
- Long

Route

A route was created by combining pickup and drop-off Taxi Zone IDs.

Example:

237 → 236

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10. NumPy Statistical Analysis

NumPy was used to calculate statistical measures for important numerical variables.

The analysis included:

- Mean
- Median
- Minimum
- Maximum
- Variance
- Standard deviation
- Percentiles

Important variables analyzed included:

- trip_distance
- trip_duration_minutes
- passenger_count
- average_speed_mph

Trip Distance

Mean: 5.86 miles

Median: 1.67 miles

The mean is higher than the median because extreme distance values influence the average.

Therefore, the median provides a better representation of a typical trip distance.

Trip Duration

Mean: 15.03 minutes

Median: 11.7 minutes

The mean is higher than the median because unusually long trips influence the average.

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11. Time-Based Analysis

Time-based analysis was performed using:

- Pickup hour
- Day of week
- Weekday/weekend

- Hour × day activity

Trips by Hour

Busiest hour: 6:00 PM

Trips: approximately 267,949

Lowest activity hour: 4:00 AM

Trips: approximately 20,031

This shows that taxi demand varies significantly throughout the day.

Trips by Day

Day	Number of Trips
Monday	363,755
Tuesday	453,966
Wednesday	577,341
Thursday	614,208
Friday	586,853
Saturday	476,393
Sunday	402,586

Highest: Thursday — 614,208 trips

Lowest: Monday — 363,755 trips

Weekday vs Weekend

Weekday: 74.71%

Weekend: 25.29%

Therefore, the majority of recorded trips occurred on weekdays.

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12. Trip Distance and Duration Analysis

Trip Distance

Average: 5.86 miles

Median: 1.67 miles

The trip distance distribution is concentrated around shorter distances, while a small number of extreme values create a long right tail.

Therefore, the median is more representative of a typical trip than the mean.

Trip Duration

Average: 15.03 minutes

Median: 11.7 minutes

The trip duration distribution is concentrated around shorter trips, while a small number of unusually long trips create a right-skewed distribution.

Distance and Duration Relationship

The correlation between trip distance and trip duration was approximately:

0.003

This indicates almost no linear relationship in the full dataset as analyzed.

However, extreme observations affect the overall relationship.

Correlation does not imply causation.

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13. Passenger Analysis

Passenger count was analyzed to identify common passenger patterns.

The most common passenger count was:

1 passenger

Approximately **79% of non-missing passenger-count records** represented one passenger.

Passenger patterns were also compared across:

- Weekdays
- Weekends
- Trip distance
- Trip duration

Single-passenger trips dominated both weekday and weekend groups.

Small passenger-count groups were interpreted carefully because some groups contained relatively few observations.

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14. Pickup and Drop-off Location Analysis

The project analyzed:

- PULocationID
- DOLocationID

These values represent Taxi Zone IDs.

Most Common Pickup Location

Zone 161

169,977 trips

Most Common Drop-off Location

Zone 236

161,376 trips

These values are reported as Taxi Zone IDs rather than neighborhood names.

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15. Route Analysis

A route was created using:

Pickup Zone → Drop-off Zone

Most Common Route

237 → 236

24,839 trips

The top 20 routes accounted for approximately:

6.37% of all trips

This shows that although some routes are more frequent, trip activity is distributed across many different routes.

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16. Payment Type Analysis

The payment_type column was analyzed as a coded categorical variable.

Payment Type	Number of Trips
Type 1	2,444,380

Payment Type	Number of Trips
Type 0	539,922
Type 2	390,409
Type 4	76,481
Type 3	23,773
Type 5	1

Payment Type 1 represented approximately:

70.34%

of the cleaned records.

The project keeps these values as payment codes rather than assigning detailed labels without an official mapping.

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17. Vendor Analysis

The distribution of recorded vendors was analyzed.

Vendor	Number of Trips
Vendor 2	2,719,728
Vendor 1	753,661
Vendor 7	1,206
Vendor 6	371

Vendor 2 accounted for the majority of recorded trips.

However, a higher trip count does not mean that one vendor is better than another.

The analysis only describes the distribution of recorded trips.

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18. Rate Code Analysis

Rate codes were analyzed to identify the most frequently recorded rate code.

The most common non-missing rate code was:

Rate Code 1

It represented approximately:

93.92% of non-missing RatecodeID records.

Other rate codes occurred much less frequently.

Rare rate codes should be interpreted carefully because they contain relatively fewer observations.

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19. Store and Forward Analysis

The store_and_fwd_flag indicates whether the trip record was stored before being sent to the vendor.

- **N** = No
- **Y** = Yes
- **Missing** = No recorded value

Flag	Records
N	2,927,398
Y	7,646
Missing	540,149

Among the recorded values, N was overwhelmingly more common.

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20. Outlier Analysis

The Interquartile Range (IQR) method was used to investigate potential outliers.

Formula

IQR = Q3 − Q1

Lower Bound = Q1 − 1.5 × IQR

Upper Bound = Q3 + 1.5 × IQR

Outlier analysis was performed for:

- Trip distance
- Trip duration
- Average speed

The project did not automatically remove all outliers.

Instead, unusual observations were investigated to determine whether they could represent:

- Valid unusual trips
- Potential data-entry issues
- Logically impossible observations
- Records requiring further investigation

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21. Unusual and Potentially Problematic Records

The analysis identified unusual records including:

- Extremely large trip distances
- Very short durations combined with very large distances
- Very long trip durations
- Zero-distance trips
- Zero-duration trips
- Extremely high calculated speeds
- Missing values
- Unusual passenger counts

Examples of extreme distance-duration combinations included:

- 73,437.25 miles in 7 minutes
- 81,231.22 miles in 5 minutes
- 82,060.16 miles in 7 minutes
- 121,555.16 miles in 8 minutes
- 143,712.27 miles in 6 minutes

These records indicate potential data-quality issues and should be investigated further.

They were retained rather than automatically deleted.

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22. Correlation Analysis

Correlation analysis was performed using numerical variables such as:

- Passenger count
- Trip distance
- Trip duration
- Average speed

The correlation between trip distance and trip duration was approximately:

0.003

This indicates almost no linear relationship in the full dataset.

Correlation does not imply causation.

Extreme observations can also affect the observed relationship.

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23. Visualizations

The project contains more than the required minimum of 12 meaningful visualizations.

Major visualizations include:

1. Trips by hour
2. Trips by day
3. Weekday vs weekend trips
4. Hour × day demand heatmap
5. Trip distance histogram
6. Trip duration histogram
7. Passenger count distribution
8. Distance by passenger count
9. Duration by passenger count
10. Payment type distribution
11. Vendor distribution
12. Rate code distribution
13. Store-and-forward distribution
14. Distance vs duration scatter plot
15. Distance outlier boxplot

16. Duration outlier boxplot
17. Average speed analysis
18. Pickup location analysis
19. Drop-off location analysis
20. Route analysis
21. Correlation heatmap

Each major visualization was followed by an interpretation explaining the important pattern observed in the data.

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24. Key Findings

The major findings from the analysis are:

- **6 PM was the busiest pickup hour**, with approximately 267,949 trips.
- **4 AM was the lowest activity hour**, with approximately 20,031 trips.
- **Weekdays represented 74.71%** of trips.
- **Weekends represented 25.29%** of trips.
- **Thursday had the highest daily trip count**, with 614,208 trips.
- **Monday had the lowest daily trip count**, with 363,755 trips.
- The **median trip distance was 1.67 miles**.
- The **median trip duration was 11.7 minutes**.
- **One passenger was the most common passenger count**, representing approximately 79% of non-missing passenger records.
- **Zone 161 was the most common pickup location**.
- **Zone 236 was the most common drop-off location**.
- **237 → 236 was the most common route**.
- **Payment Type 1 was the most common recorded payment code**, representing approximately 70.34%.
- **Rate Code 1 was the most frequently recorded rate code**.
- The dataset contains several unusual records involving distance, duration, and calculated speed.

- Missing values and extreme observations are important data-quality considerations.

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25. Business and Operational Insights

Taxi Demand

Taxi demand is concentrated during particular hours, especially during the evening.

Weekday Activity

Weekdays account for a larger share of recorded trips than weekends.

Passenger Pattern

Single-passenger trips dominate the passenger-count distribution.

Locations

Certain pickup and drop-off Taxi Zones have higher trip activity than others.

Routes

Some routes occur more frequently, although overall trip activity is distributed across many routes.

Trip Characteristics

Most typical trips are relatively short based on the median distance and duration.

Data Quality

Extreme observations can significantly affect averages and calculated speeds. Therefore, they should be investigated before being used for operational decision-making.

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26. Data Quality Limitations

The analysis has the following limitations:

1. Several columns contain substantial missing values.
2. Missing values were investigated rather than automatically deleted.
3. Taxi Zone IDs cannot be interpreted as neighborhood names without a lookup table.
4. The dataset covers only January 2025, so the results should not automatically be generalized to other months or seasons.
5. Extreme observations can strongly affect means, averages, and calculated speeds.
6. Correlation does not imply causation.

7. Very small groups and rare codes should be interpreted carefully.
8. The dataset does not provide enough information to claim:
    - Profit
    - Driver earnings
    - Customer spending
    - Customer-level behavior
    - Driver performance
9. A higher trip count for a vendor does not mean that the vendor is better.
10. Unusual records should be investigated further before being used for important operational decisions.

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27. Final KPI Summary

KPI	Value
Original Trip Records	3,475,226
Final Cleaned Records	3,475,102
Records Removed	124
Average Trip Distance	5.86 miles
Median Trip Distance	1.67 miles
Average Trip Duration	15.03 minutes
Median Trip Duration	11.7 minutes
Maximum Trip Distance	276,423.57 miles
Maximum Trip Duration	5,626.32 minutes
Most Common Passenger Count	1
Busiest Pickup Hour	6:00 PM
Lowest Activity Hour	4:00 AM
Busiest Day	Thursday

KPI	Value
Most Common Payment Code	Type 1
Most Common Pickup Location	Zone 161
Most Common Drop-off Location	Zone 236
Most Common Route	237 → 236
Weekday Share	74.71%
Weekend Share	25.29%
Distance-Duration Correlation	0.003

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28. Final Conclusion

The **NYC Yellow Taxi Trip Data Analysis** demonstrates a complete data-analysis workflow using Python, NumPy, Pandas, Matplotlib, and Seaborn.

The project began with understanding the raw dataset and assessing its quality. Missing values, duplicates, suspicious values, invalid timestamps, zero values, and extreme observations were investigated.

The data was cleaned by removing **124 records where the drop-off time occurred before the pickup time**, resulting in **3,475,102 cleaned records**.

Feature engineering was then performed to create useful variables such as trip duration, pickup hour, day of week, weekend indicator, average speed, trip categories, and routes.

Exploratory Data Analysis was performed to understand taxi demand, trip distance, trip duration, passenger patterns, locations, routes, payment types, vendors, rate codes, and relationships between numerical variables.

The analysis identified important patterns such as **6 PM being the busiest hour, weekdays accounting for 74.71% of trips, one passenger being the most common passenger count, Zone 161 being the most common pickup location, Zone 236 being the most common drop-off location, and 237 → 236 being the most common route**.

The analysis also identified unusual records involving extremely large distances, unusually short or long durations, and extreme calculated speeds. These observations demonstrate the importance of investigating data quality before making operational decisions.

Overall, this project provides a structured and evidence-based analysis of **January 2025 NYC Yellow Taxi trip activity**, while clearly documenting the limitations and unusual characteristics of the dataset.